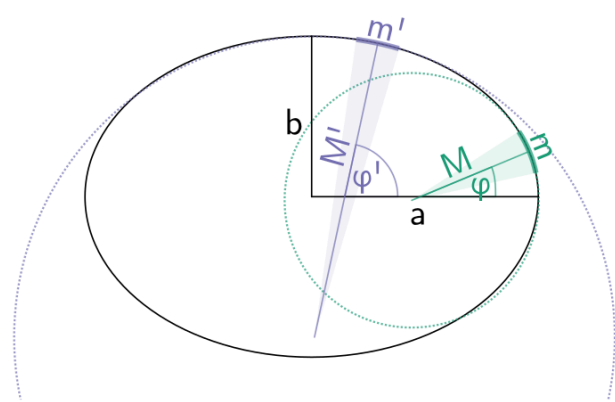


L4: Earth Ellipsoid, ellipsoidal coodimates, and satellite coordinate systems

Historic Arc measurement



meridional radius
of curvature

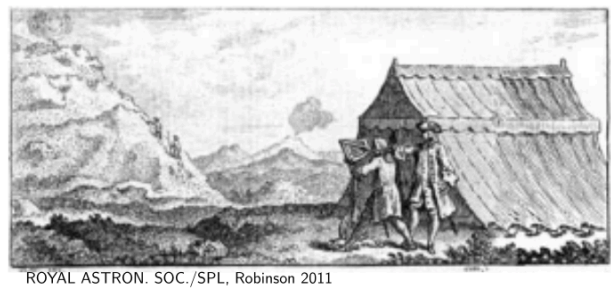
semimajor axis

$$M = \frac{a(1 - e^2)}{(1 - e^2 \sin^2(\varphi))^{\frac{3}{2}}}$$

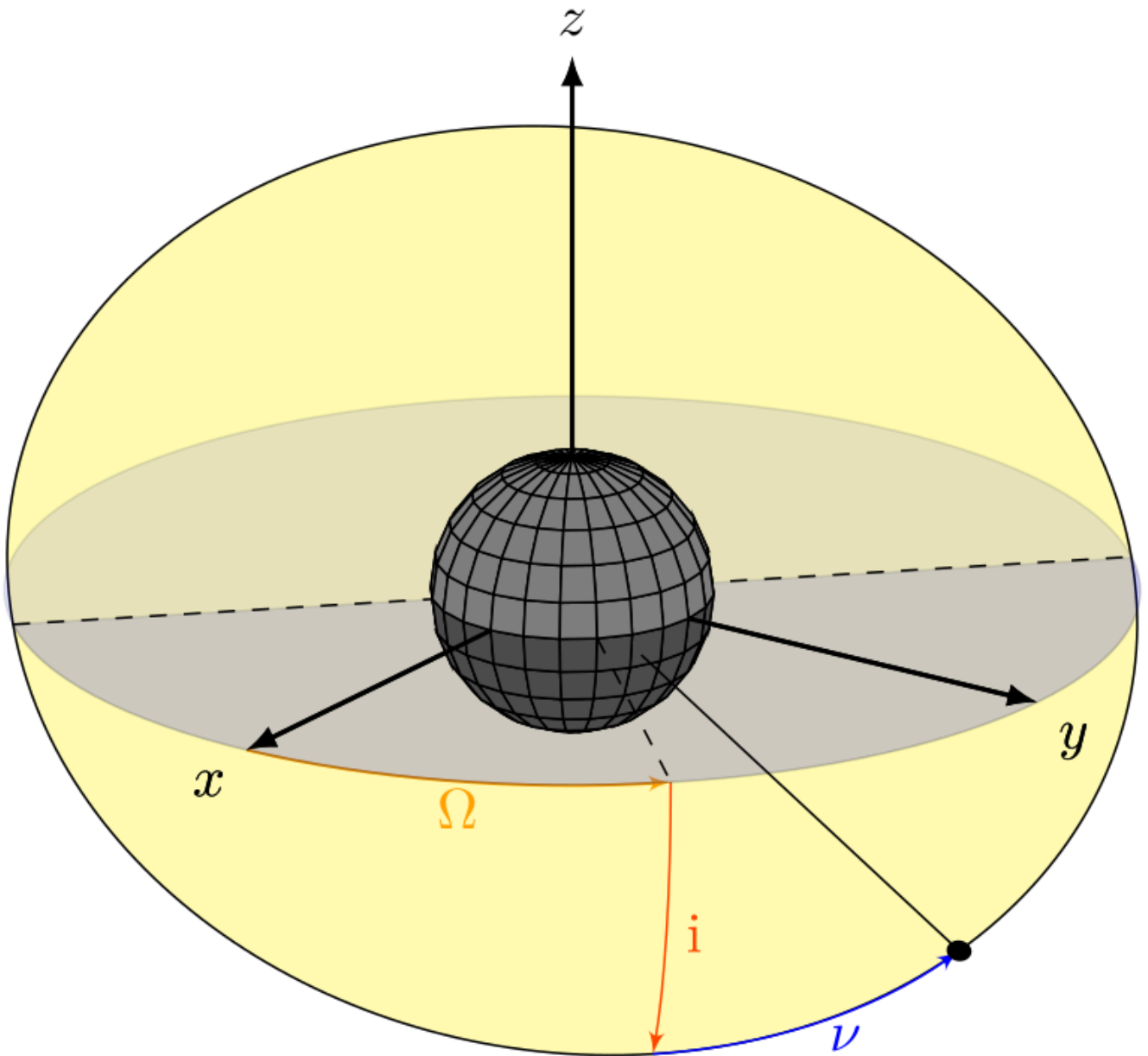
1st eccentricity

ellipsoidal latitude

Location	Latitude	Arc length (toises)
Quito	0°0'	56.751
Cape of Good Hope	33°18'	57.037
Rome	42°59'	56.979
Paris	49°23'	57.074
Lapland	66°19'	57.422



Precession of Orbital Plane



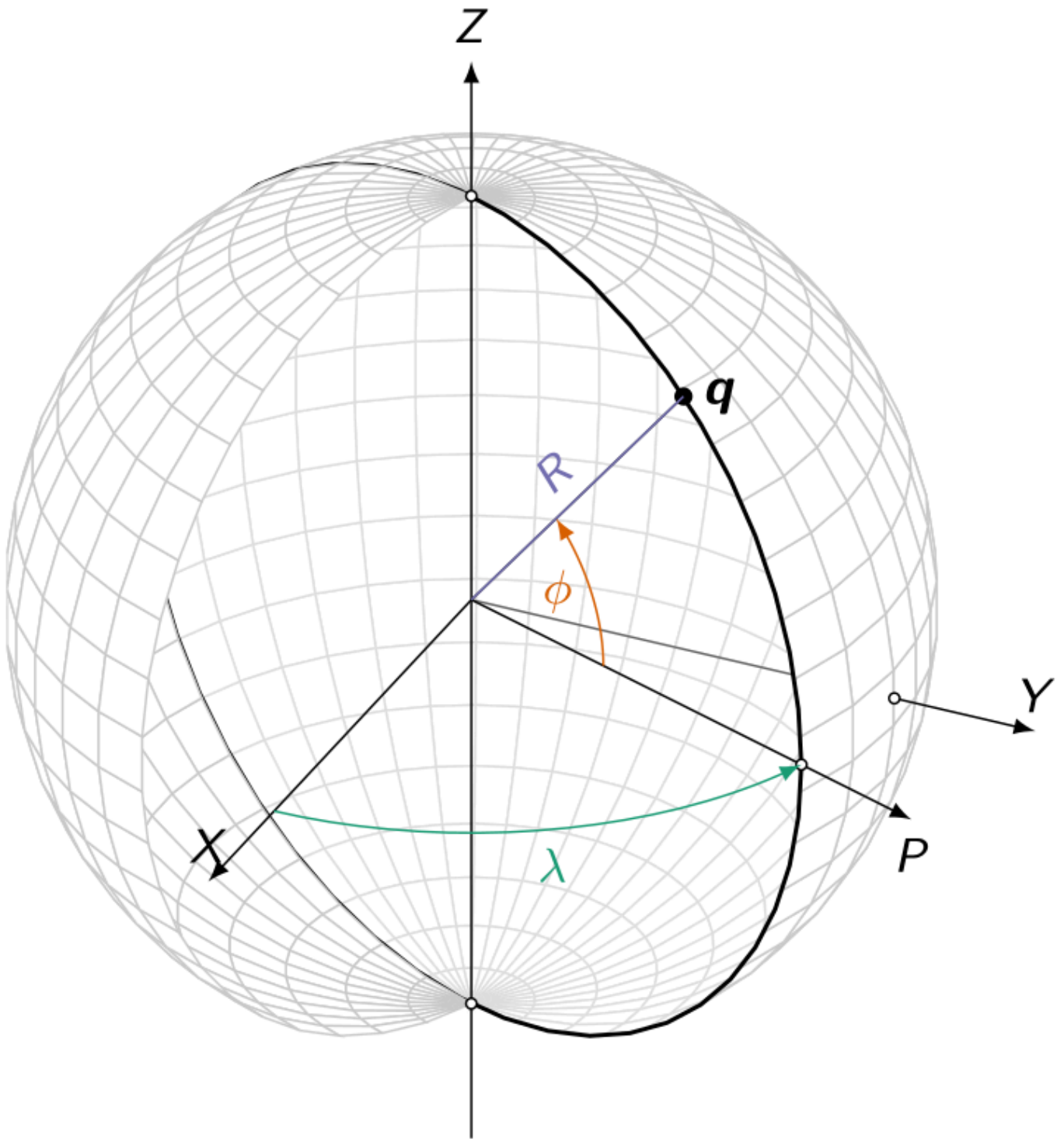
$$\dot{\Omega} = \frac{d\Omega}{dt} \neq 0$$

- From orbital theory: rotation rate of $\frac{d\Omega}{dt} = 3.2^\circ d^{-1}$ predicted
 - Confirmation by Sputnik 1 (1957)
- $\exists$ Historic measurements of Earth oblateness

Requirements

- Reference surfaces in geodesy require a definition precise at mm-level

Ellipsoidal coordinates



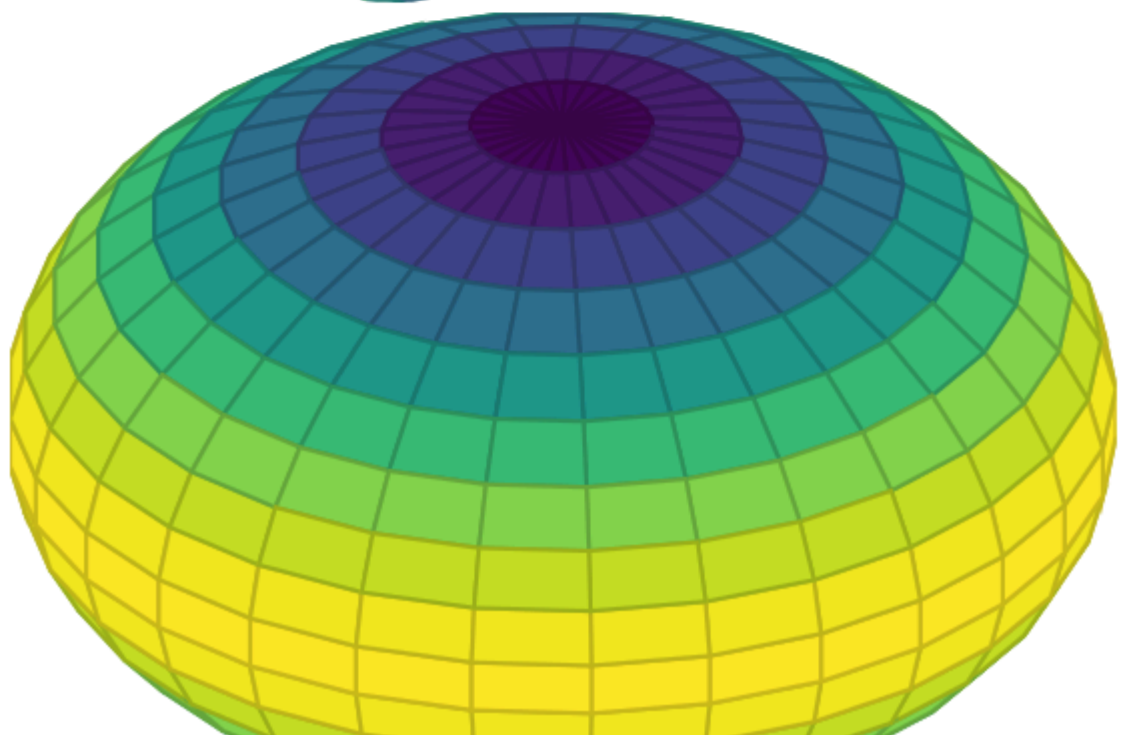
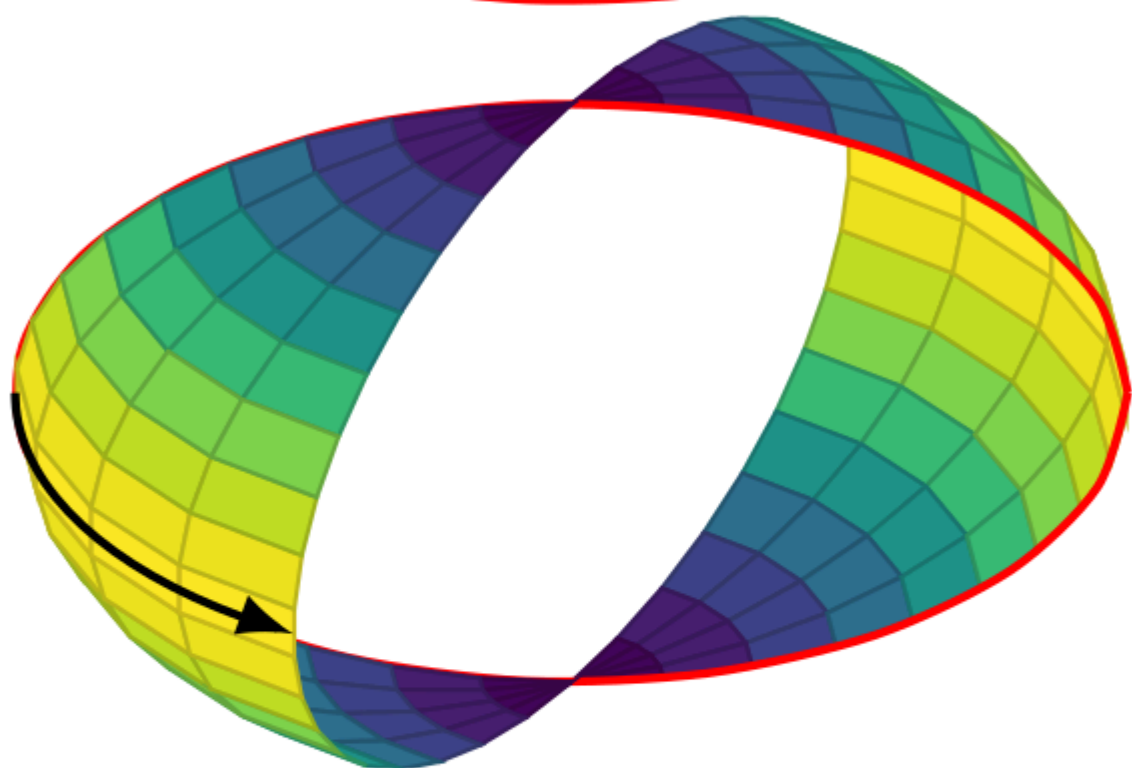
Points on Sphere

$$\mathbf{q}(\lambda, \phi) = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} R \cos \lambda \cos \phi \\ R \sin \lambda \cos \phi \\ R \sin \phi \end{bmatrix}$$

$$\lambda \in [0^\circ, 360^\circ)$$

$$\phi \in [-90^\circ, 90^\circ]$$

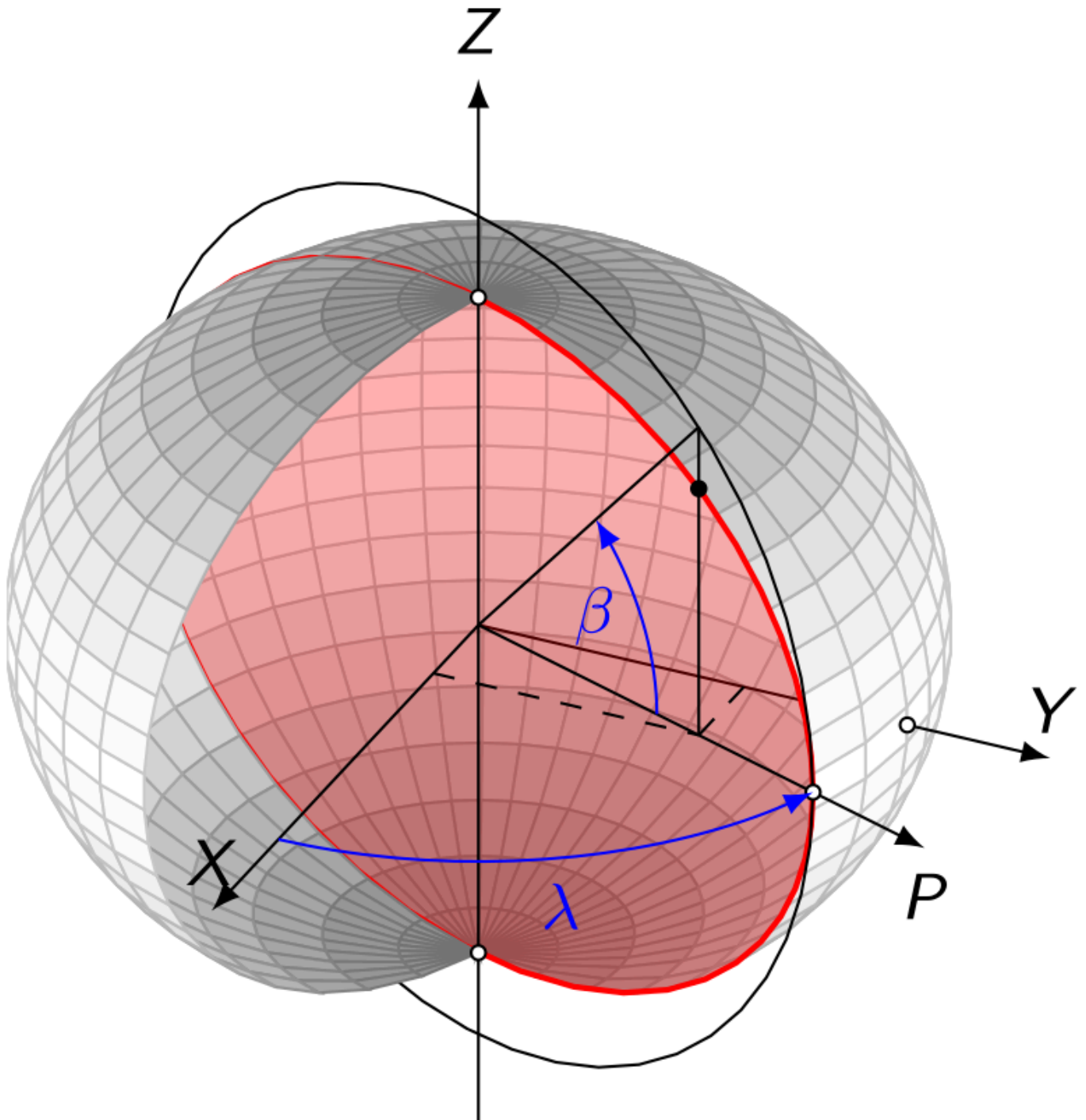
- For possible parametric representation (RHS)
- ϕ becomes β
- β parametric/reduced latitude, hard to interpret geometrically





P-Axis

- Exploit that ellipsoid of revolution is rotational symmetric around z-axis by construction

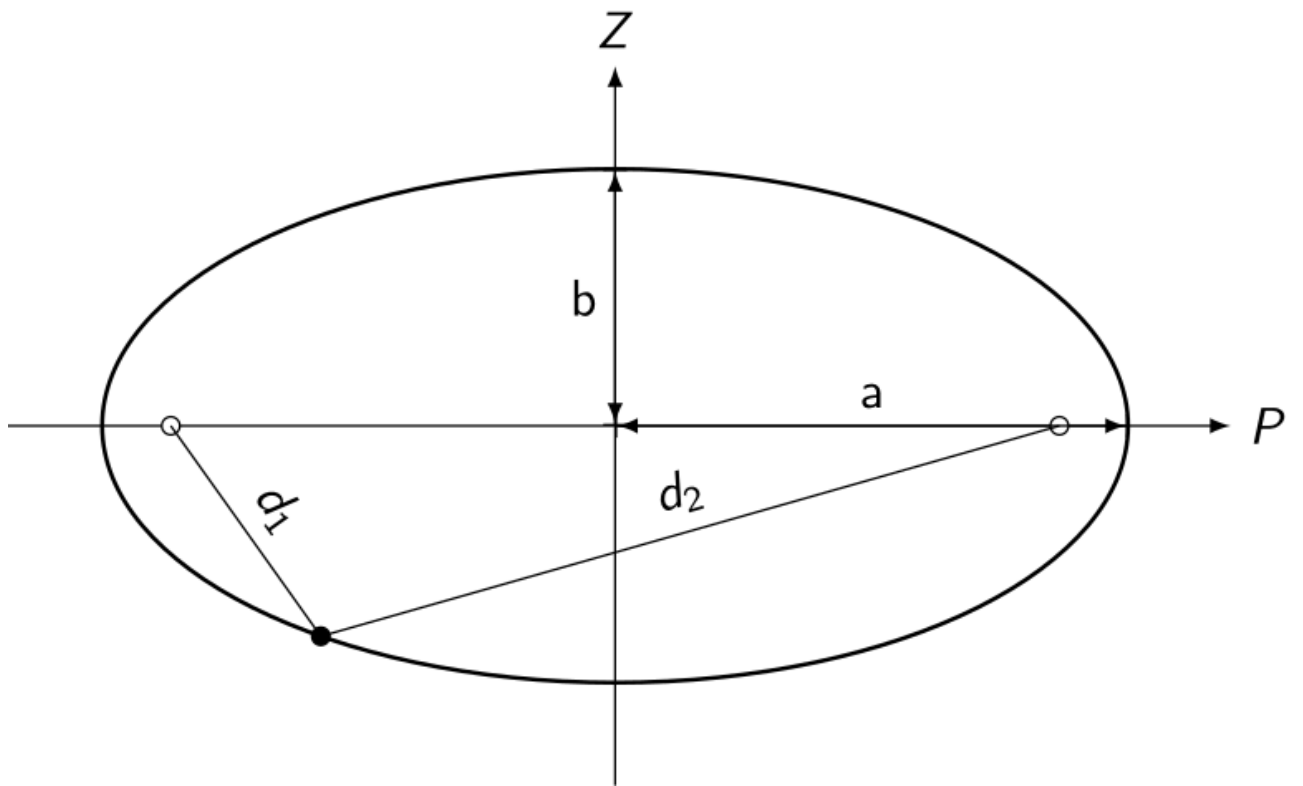


Define 'p-coordinates'

$$p = \sqrt{x^2 + y^2}$$

Ellipse

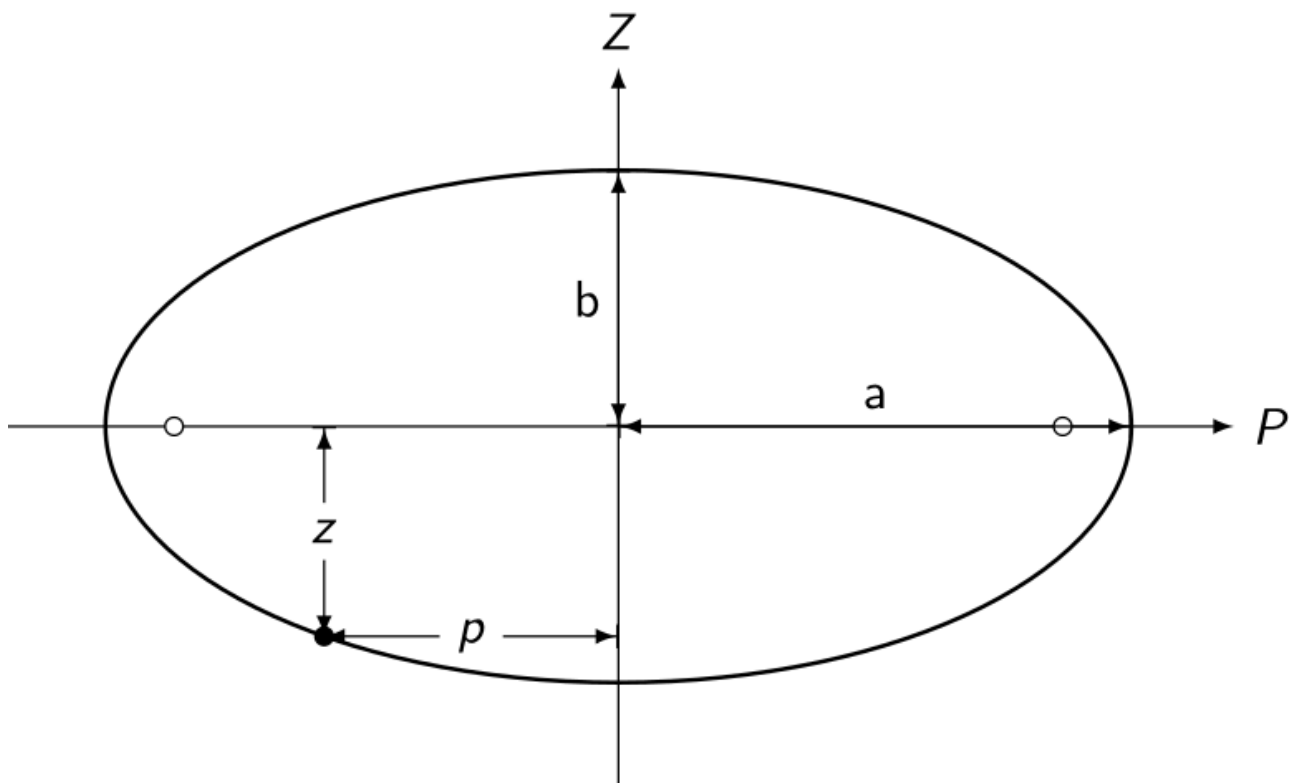
1. $d_1 + d_2 = \text{const} = 2a$



◦

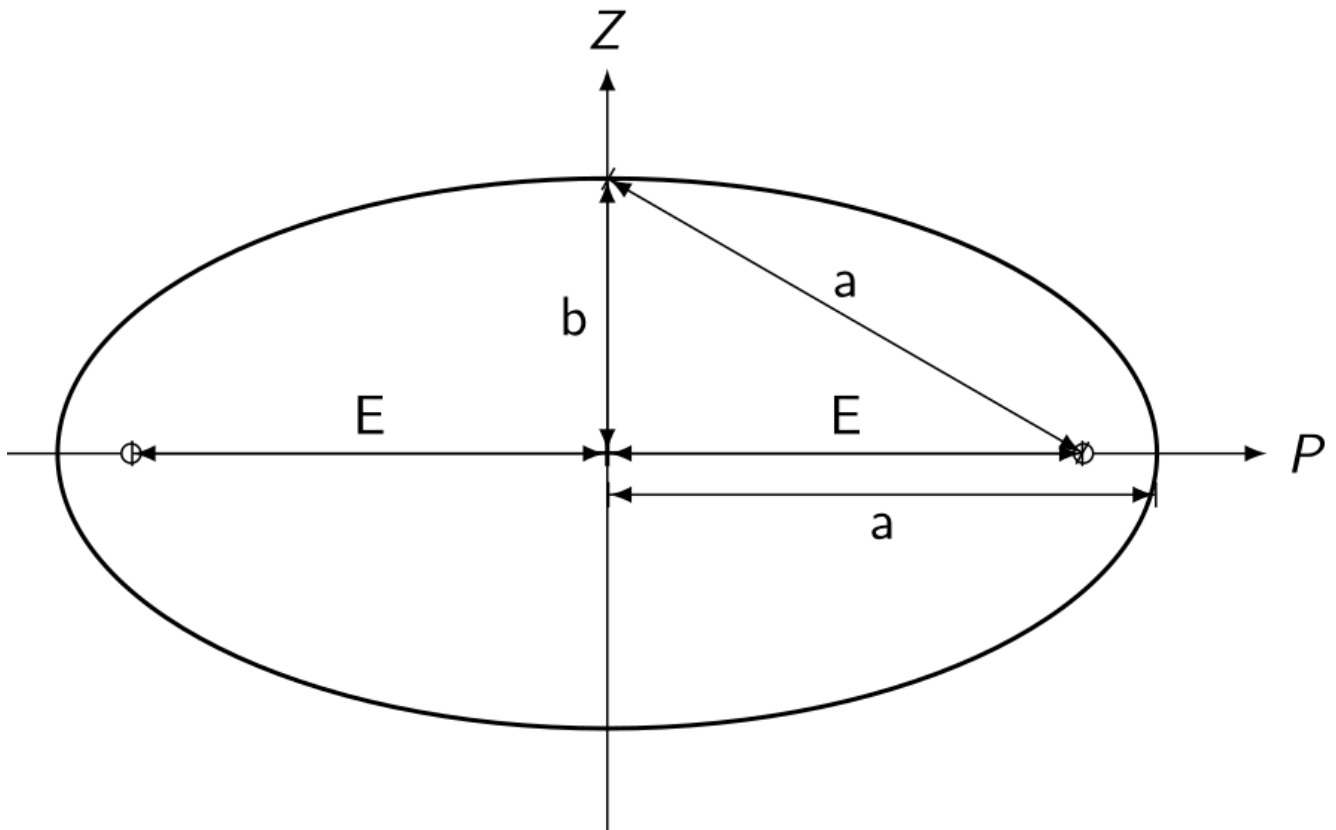
2. Equation of Ellipse:

◦ $\left(\frac{p}{a}\right)^2 + \left(\frac{z}{b}\right)^2 = 1$



◦

3. Basic laws:

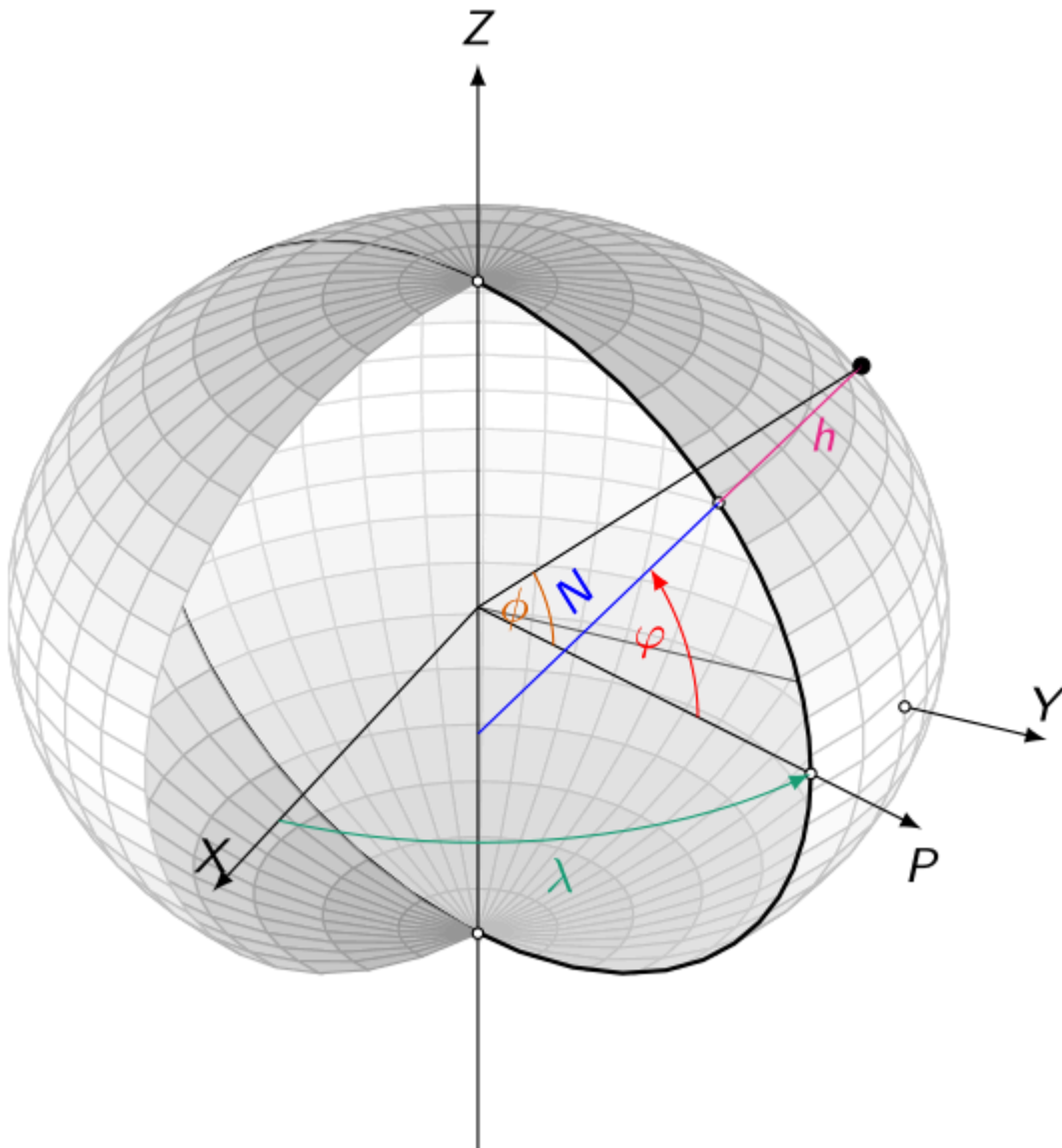


Name	Parameter	Earth
Semi-Major Axis	a	6378.137 km
Semi-Minor Axis	b	6356.752 km
Focal distance	$E = \sqrt{a^2 - b^2} = ea$	521.854 km
First eccentricity	$e = \frac{E}{a}$	0.0818 ...
Second eccentricity	$e' = \frac{E}{b}$	0.0821 ...
Oblateness	$f = (a - b)/a$	0.003 352 ..

Numerical Precision

- For given ellipsoid: 2 parameters independent
 - all other derivable
- At Maximum precision: Transformation of coordinates with 0.1 mm accuracy
 - Especially for e, e', f

Parametrisation with ellipsoidal latitude



$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2(\varphi)}}$$

$$\mathbf{q}(\lambda, \varphi) = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} (N(\varphi) + h) \cos(\lambda) \cos(\varphi) \\ (N(\varphi) + h) \sin(\lambda) \cos(\varphi) \\ (\frac{N(\varphi)}{1+e^2} + h) \sin(\varphi) \end{bmatrix}$$

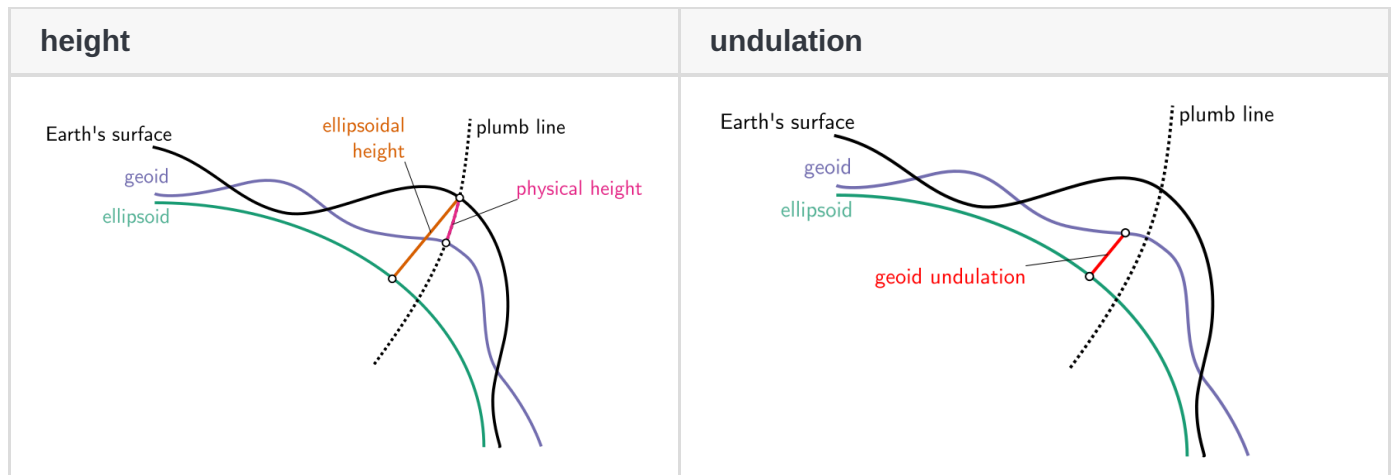
$$\lambda \in [0^\circ, 360^\circ)$$

$$\varphi \in [-90^\circ, 90^\circ]$$

Ellipsoidal coordinates alone $[\lambda, \varphi, h]$ are useless -> Requirement for:

- Geodetic datum
 - Defines origin, orientation and scale of CS
 - relative to Earth or any other celestial body
- Shape parameters of the ellipsoid
- Location and orientation of ellipsoid

Ellipsoidal and Physical Heights



Coordinate Transformations

Ellipsoidal to cartesian

1. Compute transverse radius of curvature:

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2(\varphi)}}$$

2. Compute x,y,z

$$\mathbf{q}(\lambda, \varphi) = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} (N(\varphi) + h) \cos(\lambda) \cos(\varphi) \\ (N(\varphi) + h) \sin(\lambda) \cos(\varphi) \\ (\frac{N(\varphi)}{1 + e'^2} + h) \sin(\varphi) \end{bmatrix}$$

$$\lambda \in [0^\circ, 360^\circ]$$

$$\varphi \in [-90^\circ, 90^\circ]$$

Cartesian to Ellipsoidal Coordinates

Require: x, y, z, a, b

$$\lambda \leftarrow \arctan\left(\frac{y}{x}\right)$$

$$e' \leftarrow \sqrt{\frac{a^2 - b^2}{b^2}}$$

$$e \leftarrow \sqrt{\frac{a^2 - b^2}{a^2}}$$

$$p \leftarrow \sqrt{x^2 + y^2}$$

$$h \leftarrow 0$$

▷ first guess for height

$$\varphi \leftarrow \arctan\left(\frac{z(1 + e'^2)}{p}\right)$$

▷ first guess for latitude

$$\varphi' \leftarrow \infty$$

▷ latitude from previous iteration

while $|\varphi' - \varphi| > \epsilon$ **do** ▷ choose threshold according to required precision

$$\varphi' \leftarrow \varphi$$

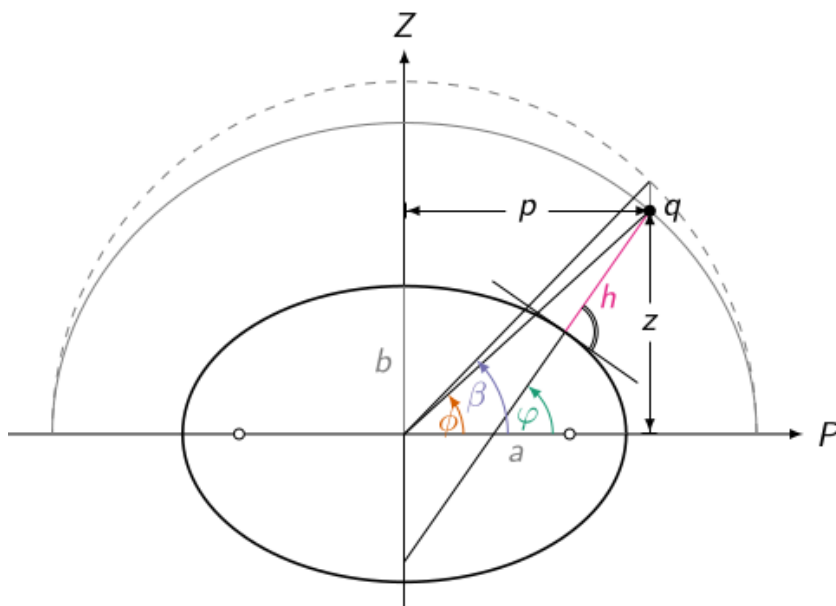
$$N \leftarrow \frac{a}{\sqrt{1 - e^2 \sin^2(\varphi)}}$$

$$h \leftarrow \frac{p}{\cos(\varphi)} - N$$

$$\varphi \leftarrow \arctan\left(\frac{z}{p} \frac{N(\varphi) + h}{\frac{N(\varphi)}{1 + e'^2} + h}\right)$$

end while

Geocentric vs. Ellipsoidal Latitude



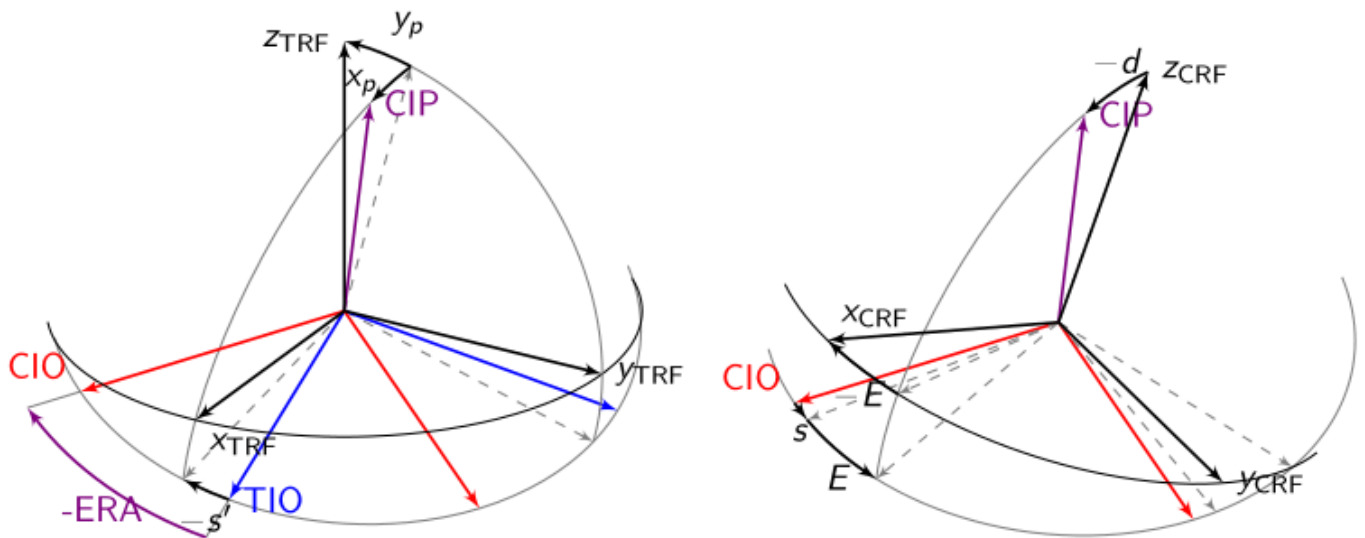
$$\phi = \arctan\left(\frac{(1 - e^2)N(\varphi)}{N(\varphi) + h} \tan(\varphi)\right)$$

Earth's rotation and inertial coordinate system

- Inertial system
 - Force-free objects remain in rest or move with uniform motion

- Also goes for constant rotation $\mathbf{R}$
- (Quasi) Inertial reference system
 - Kepler Orbit
 - Practical realisation: GCRF (Geocentric Celestial Reference Frame)
- Terrestrial reference system (co-rotating with Earth)
 - All terrestrial geodetic markers are located in
 - Reference ellipsoid refers to (z-axis for rotation)

$$\mathbf{x}_{\text{GCRS}} = \underbrace{\mathbf{R}_3(-E)\mathbf{R}_2(-d)\mathbf{R}_3(E)\mathbf{R}_3(s)}_{\text{precession and nutation } Q(t)} \underbrace{\mathbf{R}_3(-ERA)}_{\text{daily rotation } R(t)} \underbrace{\mathbf{R}_3(-s')\mathbf{R}_2(x_p)\mathbf{R}_1(y_p)}_{\text{polar motion } W(t)} \mathbf{x}_{\text{ITRS}}$$



For this lecture:

$$\mathbf{R} = \mathbf{R}_3(\omega_{\oplus}(t - t_0))$$

Satellite ground tracks

Mean Anomaly M

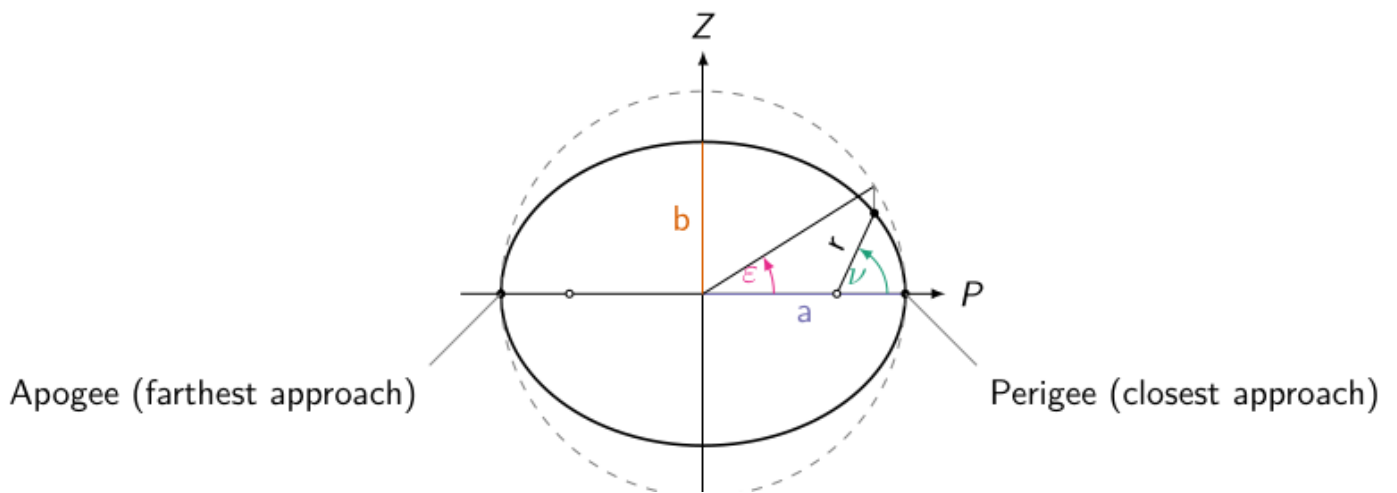
$$M(t) = \frac{2\pi}{T}(t - t_0) = \sqrt{\frac{GM}{a^3}}(t - t_0)$$

Kepler Equation

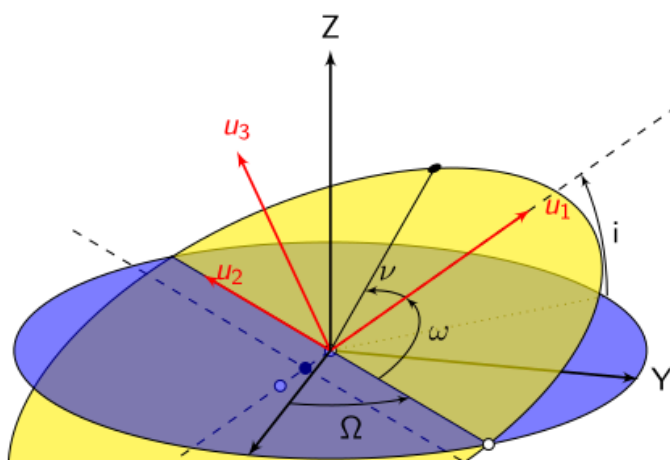
$$M(t) = \underbrace{\varepsilon(t)}_{\text{Eccentric anomaly}} - e \sin(\varepsilon(t))$$

True anomaly ν

$$\cos(\nu(t)) = \frac{\cos(\varepsilon(t)) - e}{1 - e \cos(\varepsilon(t))}$$



Transformation: Inertial to orbital coordinate system

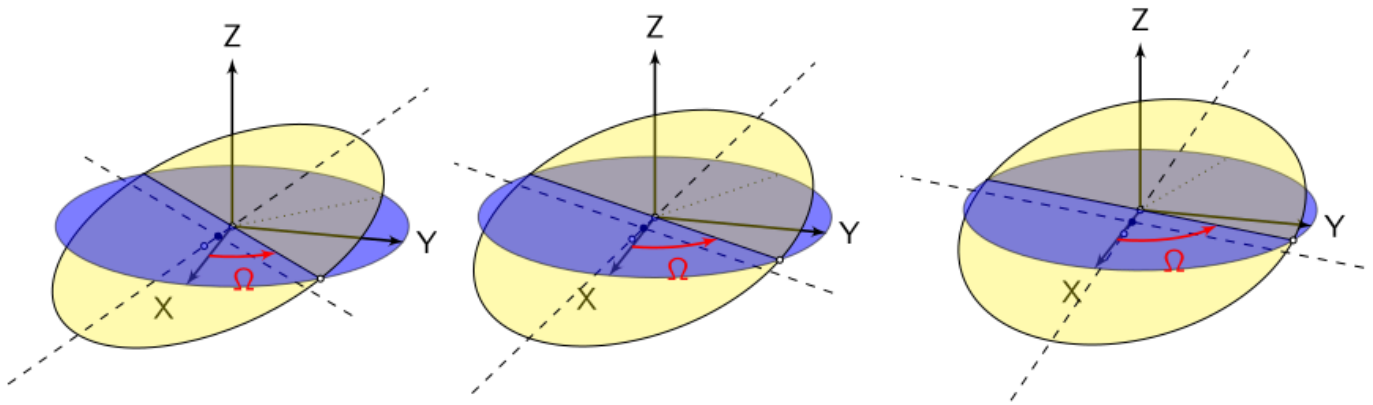


$$R = R_3(\omega)R_1(i)R_3(\Omega)$$

$$R = \begin{bmatrix} \cos(\Omega) \cos(\omega) - \sin(\Omega) \sin(\omega) \cos(i) & -\cos(\Omega) \sin(\omega) - \sin(\Omega) \cos(\omega) \cos(i) & \sin(\Omega) \sin(i) \\ \sin(\Omega) \cos(\omega) + \cos(\Omega) \sin(\omega) \cos(i) & -\sin(\Omega) \sin(\omega) + \cos(\Omega) \cos(\omega) \cos(i) & -\cos(\Omega) \sin(i) \\ \sin(\omega) \sin(i) & \cos(\omega) \sin(i) & \cos(i) \end{bmatrix}$$

Orbital Precession

- Kepler Problem assumes point masses (equivalent to spheres with homogeneous density distribution) -> isotropic gravity field
- But: Earth flattens
- -> Ω changes with time
- "Orbital precession"



Orbitl Precession Rate

Orbital precession rate $\dot{\Omega}$

mean motion

Earth radius

dynamical flattening

$$\dot{\Omega} = -\frac{3R^2 n J_2}{2(1-e^2)^2 a^2} \cos(i)$$

1st eccentricity

semimajor axis

inclination

Orbital precession in Earth fixed frame

$$\dot{\Omega}_e = \dot{\Omega} - \omega_{\oplus} = -\frac{3R^2 n J_2}{2(1-e^2)^2 a^2} \cos(i) - \omega_{\oplus}$$

$\omega_{\oplus}$: Earth rotation rate (do not confuse with argument of perigee)

Approximating Ground Tracks

- For Visualisation:
- Transformation from Keplerian elements to geographical coordinates of sub-satellite point, assuming
 - Circular orbit ($E = e = e' = 0$)
 - Ω, i, ω are constant (point masses)
 - Position along orbit: mean anomaly
- Not accurate at cm-level

$$\phi = \arcsin(\sin(i) \sin(\omega + \nu))$$

$$\lambda = \omega_{\oplus}(t - t_0) + \Omega + \arctan\left(\frac{\cos(\omega + \nu) \tan(i)}{\sin(\phi)}\right)$$

with $\omega_{\oplus} = \frac{360^\circ}{\text{sidereal day}}$